MTH2003 Feedback Sheet 2 Solutions

1. (***a***) **Question** $y'' + 8y' + 25y = 800t\cos(5t)$, $y(0) = 4$, $y'(0) = -16$.

Solution

[i] The complementary solution is obtained by using the characteristic equation for the corresponding homogeneous problem: $m^2 + 8m + 25 = 0$ yielding $m = -4 \pm 3\mathrm{i}$ and so

$$\boxed{y = \mathrm{e}^{-4t}[A\cos(3t) + B\sin(3t)]}.$$

[ii] We look for a particular solution in the form $y = (\alpha t + \beta)\cos(5t) + (\gamma t + \delta)\sin(5t)$. Then $\dfrac{\mathrm{d}y}{\mathrm{d}t} = \alpha\cos(5t) + \gamma\sin(5t) - 5(\alpha t + \beta)\sin(5t) + 5(\gamma t + \delta)\cos(5t) = (\alpha + 5\gamma t + 5\delta)\cos(5t) + (\gamma - 5\alpha t - 5\beta)\sin(5t)$ and $\dfrac{\mathrm{d}^2y}{\mathrm{d}t^2} = [10\gamma - 25\alpha t - 25\beta]\cos(5t) + [-25\gamma t - 25\delta - 10\alpha]\sin(5t)$. Substituting these into the differential equation gives, after collecting similar terms:

$$[8\alpha + 10\gamma + 40\delta + 40\gamma t]\cos(5t) + [-10\alpha - 40\beta + 8\gamma - 40\alpha t]\sin(5t) = 800t\cos(5t).$$

Firstly, we equate coefficients of $t\cos(5t)$ on both sides of the equation. We find $40\gamma = 800$ so $\gamma = 20$. We then equate coefficients of $t\sin(5t)$ on both sides of the equation. We find $-40\alpha = 0$. We now equate coefficients of $\cos(5t)$ on both sides of the equation. We find $8\alpha + 10\gamma + 40\delta = 0$ and since $\alpha = 0$ and $\gamma = 20$, we have $\delta = -5$. Finally, we equate coefficients of $\sin(5t)$ on both sides. We get $-10\alpha - 40\beta + 8\gamma = 0$ and since $\alpha = 0$ and $\gamma = 20$, we have $\beta = 4$. Substituting these values into our trial solution, our particular solution $y = (\alpha t + \beta)\cos(5t) + (\gamma t + \delta)\sin(5t)$, becomes

$$\boxed{y = 4\cos(5t) + (20t - 5)\sin(5t)}.$$

[iii] The general solution is therefore:

$$\boxed{y = \mathrm{e}^{-4t}\left(A\cos(3t) + B\sin(3t)\right) + 4\cos(5t) + (20t - 5)\sin(5t)}$$

[iv] From the first initial condition, $y(0) = 4 = A + 4$, so we see that $A = 0$. Therefore, $y = B\mathrm{e}^{-4t}\sin(3t) + 4\cos(5t) + (20t - 5)\sin(5t)$ and differentiating gives $y' = \mathrm{e}^{-4t}\left(3B\cos(3t) - 4B\sin(3t)\right) + (100t - 25)\cos(5t)$. The second condition then gives $y'(0) = -16 = 3B - 25$, so $B = 3$. The solution is therefore

$$\boxed{y = 3\mathrm{e}^{-4t}\sin(3t) + 4\cos(5t) + (20t - 5)\sin(5t)}$$

2. (***a***) **Question** $(1-t^2)\dfrac{\mathrm{d}^2y}{\mathrm{d}t^2}+2t\dfrac{\mathrm{d}y}{\mathrm{d}t}-2y=0 \quad (t\neq\pm1), \quad y_1=t.$

Solution

[i] If $y=t$, $\mathrm{d}y/\mathrm{d}t=1$, $\mathrm{d}^2y/\mathrm{d}t^2=0$, so the left-hand side of the equation gives $(1-t^2)\cdot 0+2t\cdot 1-2\cdot t=2t-2t=0$, which is the right-hand side. Thus $y=t$ is a solution.

[ii] We now substitute $y=vt$. This gives

$$(1-t^2)\left(t\frac{\mathrm{d}^2v}{\mathrm{d}t^2}+2\frac{\mathrm{d}v}{\mathrm{d}t}\right)+2t\left(t\frac{\mathrm{d}v}{\mathrm{d}t}+v\right)-2tv=0.$$

We see that the two terms in v cancel leaving us with

$$(1-t^2)\left(t\frac{\mathrm{d}^2v}{\mathrm{d}t^2}+2\frac{\mathrm{d}v}{\mathrm{d}t}\right)+2t^2\frac{\mathrm{d}v}{\mathrm{d}t}=0.$$

This equation does not involve v itself. It is in fact a first order differential equation for v'. If we write $w=v'$ the equation simplifies to

$$t(1-t^2)\frac{\mathrm{d}w}{\mathrm{d}t}+2w=0$$

and dividing through by $(1-t^2)t$ gives, for $t\neq 0,\pm1$,

$$\frac{\mathrm{d}w}{\mathrm{d}t}+\left(\frac{2}{t}+\frac{2t}{1-t^2}\right)w=0.$$

This can be solved by separating the variables. First, note that $w\equiv 0$ is the special solution. For $w\neq 0$, we have $\int\frac{dw}{w}=-\int\left(\frac{2}{t}+\frac{2t}{1-t^2}\right)\mathrm{d}t \Rightarrow \ln|w|=-2\ln|t|+\ln|1-t^2|+C$. Therefore $w=A(1-t^2)/t^2=\mathrm{d}v/\mathrm{d}t$, where $A=\pm\mathrm{e}^C$, and the special solution is covered by $A=0$. Now we integrate and find $v=A(-1/t-t)+B$. Thus the solution for $y=-A(1+t^2)+Bt$. The first solution was $y_1=t$. So the second solution can be chosen with $A=-1$ and $B=0$, for example, which gives $y_2=1+t^2$.

[iii] If $y=1+t^2$, then $y'=2t$ and $y''=2$. Substituting this into the left-hand side of the ODE for y we have

$$(1-t^2)(2)+2t(2t)-2(1+t^2)=2-2t^2+4t^2-2-2t^2=0=RHS$$

[iv] We have $W[y_1,y_2]=\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}=\begin{vmatrix} t & t^2+1 \\ 1 & 2t \end{vmatrix}=t^2-1\neq 0$ since by assumption $t\neq\pm1$ (these are in fact singular points of the ODE, i.e. coefficients $p(t)$ and $q(t)$ in the normalised form of the ODE are not defined at these points) hence y_1 and y_2 are linearly independent.

3. (***a***) **Question** $t^2y'' - 4ty' + 6y = 21t^{-4}$, $t \neq 0$.

Solution The homogeneous equation $t^2y'' - 4ty' + 6y = 0$ was considered on the Examples Sheet Q2d so we know that $y_1 = t^2$, $y_2 = t^3$. Their Wronskian is $W = \begin{vmatrix} t^2 & t^3 \\ 2t & 3t^2 \end{vmatrix} = 3t^2t^2 - t^32t = t^4$. Before using the formula for the solution of the nonhomogeneous equation, we should bring the equation to the normalised form, which is $y'' - 4t^{-1}y' + 6t^{-2}y = 21t^{-6}$, so $p = -4t^{-1}$, $q = 6t^{-2}$ and $g = 21t^{-6}$. Substituting $g(t)$ and $W(t)$ into the formula, we get

$$y = -t^2 \int \frac{t^3 \cdot 21t^{-6}}{t^4}\,\mathrm{d}t + t^3 \int \frac{t^2 \cdot 21t^{-6}}{t^4}\,\mathrm{d}t$$

$$= -21t^2 \int t^{-7}\,\mathrm{d}t + 21t^3 \int t^{-8}\,\mathrm{d}t$$

$$= -21t^2 \frac{t^{-6}}{(-6)} + 21t^3 \frac{t^{-7}}{(-7)}$$

$$= \frac{7}{2}t^{-4} - 3t^{-4} = \frac{1}{2}t^{-4} \text{ for a particular integral, and the general solution}$$

$$\boxed{y = At^2 + Bt^3 + \frac{1}{2}t^{-4}.}$$